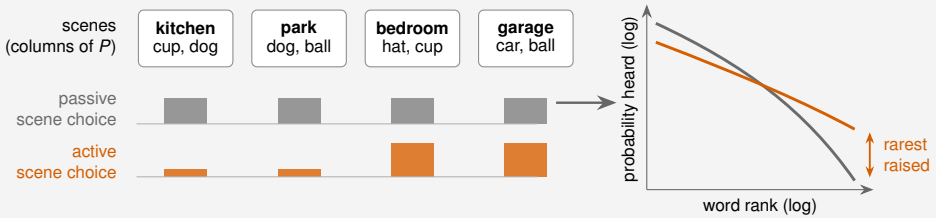
A Static control: choosing scenes (fast-mapping analysis)



Passive: scenes at random, so the rarest words (car, hat) are heard very rarely.

Active: mix scenes to make the rarest word as likely as possible, $\hat{q} = \arg\max_q \min(Pq)$, or estimate P online.

235× faster on SUN scenes, 6.7× on Flickr8k; online matches optimal.

B Adaptive control: choosing the target (incremental learners)

Learner's state

ball ✓ cup ✓

dog ? car ? hat ?

Each unlearned word keeps a set of candidate referents, e.g. "car": {car, cup, dog}

A word is learned once one candidate remains

1. Which word?

Passive: drawn by frequency, e.g. "ball" again, although already known; $C - 1$ distractors drawn by frequency

Active: a word not yet known, e.g. "car"; same random distractors

2–10× faster at 50–90% of the vocabulary; 19–28× at 99% (1,000–60,000 words)

2. Update

Eliminative: drop candidates absent from this scene

Guess-test: keep one guess; drop it if absent from this scene

Rare words are the passive learner's bottleneck

Under forgetting, returning to not-yet-learned words doubles as rehearsal: the active/passive gap at 99% widens from 4.4× to 467× as memory gets less durable